

Nan Li

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EDUCATIONAL BACKGROUND

Georgia Institute of Technology Ph.D in Robotics Research Interest: Multi-robot Collaborate Transport, World Model, Loco-Manipulation	Atlanta, United States May.2026-Jun.2030 (Expected)
Georgia Institute of Technology M.S in Mechanical Engineering Cumulative GPA: 4.0/4.0 Core Courses: Linear Control, Adv Control Design Implement, Intro to Robotics Research, Machine Learning, Computer Vision	Atlanta, United States Aug.2024-Jun.2026
Sichuan University B.Eng in Mechanical Engineering Cumulative GPA: 3.93/4.0 Rank: 3/81 TOEFL: 103 Core Courses: Dynamics Systems, Automatic Controls, Mechanical Design, Linear Algebra, Probability	Chengdu, China Sep.2020-Jun.2024

PUBLICATIONS AND WORKSHOPS

Publications

- [1] Nan Li, Jiming Ren, Haris Miller, Samuel Coogan, Karen M. Feigh, and Ye Zhao, “**Adaptive Obstacle-Aware Task Assignment and Planning for Heterogeneous Robot Teaming**”, Accepted by *IEEE Transactions on Automation Science and Engineering (T-ASE)* [\[pdf\]](#)
- [2] Xinyi Liu, Nan Li* (Co-First Author), Yifan Wang, Yuanye Dong, Beining Fu, and Qi Lu, “**Singularity Free Dynamic Control Allocation for a Tilt-rotor Multirotor Unmanned Aerial Vehicle**”, Accepted by *IEEE CASE 2023* [\[pdf\]](#)
- [3] Fashu Xu, Wenjun Huang, Hao He, Nan Li, Kang Li, and Hongchen He, “**A Segmented Dynamic Movement Primitives-Based Gait Assistive Strategy for Soft Ankle Exosuit**”, Accepted by *ICIRA 2023* [\[pdf\]](#)

- [1] Nan Li, Haris Miller, Jiming Ren, Alagappan Swaminathan, Samuel Coogan, Karen M. Feigh, and Ye Zhao, “**Resilient Task Allocation and Planning Framework for Heterogeneous Robot Teams**”, ICRA 2025 Workshop [\[pdf\]](#)

RESEARCH EXPERIENCE

Object-centric World Model for Multi-robot Collaborate Transport Supervised by Dr. Ye Zhao and Dr. Seung-Kyum Choi Georgia Institute of Technology Core Contents: - Develop an object-centric world model to predict payload states and support coordinated decision-making in multi-robot collaborative transport. - Evaluate the proposed method on heterogeneous robot teams, including wheeled humanoids, quadrupeds, and autonomous mobile robots. - Develop a decentralized, communication-free framework for lightweight and generalizable multi-robot coordination.	May. 2026–Present
Multi-agent System Task Allocation and Path Planning Supervised by Dr. Ye Zhao Georgia Institute of Technology Core Contents: - Propose an adaptive Halton sequence map, where sampling density automatically adjusts according to obstacle distribution. - Develop a hierarchical heterogeneous allocation framework based on a cluster-auction-task selection structure, which generalizes to any number of task types and reduces allocation complexity while maintaining capacity and capability constraints. - Present a fully integrated online pipeline with LLM-guided interaction for multi-robot system. It interprets natural language inputs and supports real-time replanning in response to human instructions. - Validate the OATH framework in various case studies, showing substantial improvements in task assignment quality, scalability, adaptability to dynamic changes, and overall execution performance compared to state-of-the-art MATP baselines.	Jan. 2025–Feb. 2026
Singularity Free Dynamic Control Allocation for a Tilt-rotor Multirotor Unmanned Aerial Vehicles Supervised by Prof. Qi Lu Sichuan University Core Contents: Build the barrier function to convert the inequality constraint into an equation constraint to circumvent the singularity problem;	Jan.2022-Mar.2023

- Introduced the equation constraint via Lagrange's equation and converted this problem to an optimization problem;
- Proved the stability of the adaptive algorithm by using Barabara's Lemma;
- Performed the trajectory simulation on the six-rotor tilt-rotor UAV avoidance singularity problem.

A Segmented Dynamic Movement Primitives-Based Gait Assistive Strategy for Soft Ankle Exosuit
 Huaxi Medical Robot Research Institute | Sichuan UniversityMar.2023-Jun.2023

Core Contents:

- Learned about the different gait planning algorithms and analyzed the advantages and disadvantages of each one;
- Grasped the DMP gait planning approach and applied it to the soft exoskeleton design;
- Made the DMP gait planning adaptable to different users by using reinforcement learning and optimization algorithms.

Design and Integration of Automatic Reagent Preparation Machine
 Huaxi Medical Robot Research Institute (Leader of this project) | Sichuan UniversityAug.2023-Apr.2024

Core Contents:

- Design specific components of the reagent machine;
- Use DH parameters to model the robot arm, enabling precise calculation of its position based on input joint angles;
- Learn the use of QT Creator to develop a user-friendly human-computer interaction interface for controlling the robot arm;

HONORS & AWARDS

Academic Star (Top 1%) Sichuan University	2020
First Prize of Academic Scholarship (Top 3%) Sichuan University	2020
Merit Student (Top 5%) Sichuan University	2020, 2021, 2022
Dean's List (Top 5%) Sichuan University	2021, 2022
Second Prize of Academic Scholarship Sichuan University	2022

EXTRACURRICULAR ACTIVITIES

Graduate Assistant, Student Assistant Georgia Institute of Technology	2025 Spring-present
Senior Project Leader Sichuan University - Pittsburgh Institute	2023-2024
Teaching Assistant Course of Analytical Geometry and Calculus	2022 Fall, 2023 Spring
Teaching Assistant Course of Mechanical Design 1	2023 Fall, 2024 Spring
Teaching Assistant Course of Dynamic Systems	2023 Fall, 2024 Spring
Peer Advisor Sichuan University - Pittsburgh Institute	2021-2022
Leader of Sichuan University - Pittsburgh Institute Debate Team	2021-2022

PROFESSIONAL SERVICE

Journal Reviewer	
<i>IEEE Transactions on Robotics (T-RO)</i>	2025
<i>IEEE/ASME Transactions on Mechatronics (T-Mech)</i>	2025

Volunteer	
<i>IEEE International Conference on Robotics and Automation (ICRA)</i>	2025

PROFESSIONAL SKILLS

Programming Language: Python, MATLAB, C
Tools: ROS2, NVIDIA Isaac Sim, Catia, SolidWorks